

- (c) Sodium fluoride is added to some UK public water supplies to reduce tooth decay in children.

In America sodium hexafluorosilicate, Na_2SiF_6 , is more commonly used. The relative formula mass of sodium hexafluorosilicate is 188.

- (i) Calculate the percentage of fluorine in sodium hexafluorosilicate. [2]

$$A_r(\text{F}) = 19 \quad M_r(\text{Na}_2\text{SiF}_6) = 188$$

Percentage = %

- (ii) State an **ethical** reason why some people oppose the fluoridation of water supplies. [1]

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- (iii) Apart from water supplies, state the most commonly used source of fluoride to reduce tooth decay. [1]

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END OF PAPER



FORMULAE FOR SOME COMMON IONS

POSITIVE IONS		NEGATIVE IONS	
Name	Formula	Name	Formula
aluminium	Al^{3+}	bromide	Br^{-}
ammonium	NH_4^{+}	carbonate	CO_3^{2-}
barium	Ba^{2+}	chloride	Cl^{-}
calcium	Ca^{2+}	fluoride	F^{-}
copper(II)	Cu^{2+}	hydroxide	OH^{-}
hydrogen	H^{+}	iodide	I^{-}
iron(II)	Fe^{2+}	nitrate	NO_3^{-}
iron(III)	Fe^{3+}	oxide	O^{2-}
lithium	Li^{+}	sulfate	SO_4^{2-}
magnesium	Mg^{2+}		
nickel	Ni^{2+}		
potassium	K^{+}		
silver	Ag^{+}		
sodium	Na^{+}		
zinc	Zn^{2+}		



2
1

Group

3



6

7

0

1 H Hydrogen 1																																					
7	Li	9	Be							11	B	12	C	14	N	16	O	19	F	20	Ne																
Lithium	3	Beryllium	4							Boron	5	Carbon	6	Nitrogen	7	Oxygen	8	Fluorine	9	Neon	10																
23	Na	24	Mg							27	Al	28	Si	31	P	32	S	35.5	Cl	40	Ar																
Sodium	11	Magnesium	12							Aluminium	13	Silicon	14	Phosphorus	15	Sulfur	16	Chlorine	17	Argon	18																
39	K	40	Ca	45	Sc	48	Ti	51	V	52	Cr	55	Mn	56	Fe	59	Co	59	Ni	63.5	Cu	65	Zn	70	Ga	73	Ge	75	As	79	Se	80	Br	84	Kr		
Potassium	19	Calcium	20	Scandium	21	Titanium	22	Vanadium	23	Chromium	24	Manganese	25	Iron	26	Cobalt	27	Rhodium	45	Palladium	46	Silver	47	Cadmium	48	Gallium	31	Germanium	32	Arsenic	33	Selenium	34	Bromine	35	Krypton	36
86	Rb	88	Sr	89	Y	91	Zr	93	Nb	96	Mo	99	Tc	101	Ru	103	Rh	106	Pd	108	Ag	112	Cd	115	In	119	Sn	122	Sb	128	Te	127	I	131	Xe		
Rubidium	37	Strontium	38	Yttrium	39	Zirconium	40	Niobium	41	Molybdenum	42	Technetium	43	Ruthenium	44	Rhodium	45	Palladium	46	Silver	47	Cadmium	48	Indium	49	Tin	50	Antimony	51	Tellurium	52	Iodine	53	Xenon	54		
133	Cs	137	Ba	139	La	179	Hf	181	Ta	184	W	186	Re	190	Os	192	Ir	195	Pt	197	Au	201	Hg	204	Tl	207	Pb	209	Bi	210	Po	210	At	222	Rn		
Caesium	55	Barium	56	Lanthanum	57	Hafnium	72	Tantalum	73	Tungsten	74	Rhenium	75	Osmium	76	Iridium	77	Platinum	78	Gold	79	Mercury	80	Thallium	81	Lead	82	Bismuth	83	Polonium	84	Astatine	85	Radon	86		
223	Fr	226	Ra	227	Ac																																
Francium	87	Radium	88	Actinium	89																																

Key

Key

relative atomic mass

Symbol	Name	atomic number
Ar		Z